Mars in the Making:
Final Design Project Report

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Introduction

In the future, NASA plans to build a human colony on Mars for scientific research. As humans move to Mars, they will find life very different from living on Earth. According to the article "Could We Really Live on Mars," humans will be limited to living inside of a base unable to move freely (BBC). In other words, unlike cities on Earth that provide comfort and safety to humanity, those living on Mars will be living in scarce conditions. To alleviate the stress and isolation that comes with living on Mars in a small colony, value will be created by fostering a sense of community within those on Mars. This is significant because of the positive effects it will have on the mental state of the inhabitants. For instance, the article "How Social Isolation Affects the Brain" denotes that social isolation can decrease white matter in brain regions "critical for thinking and emotional control" (Psychology Today). Rather than allowing the inhabitants of Mars to mentally deteriorate because of social isolation within the base, causing lower functionality on the base overall, the design will bring the inhabitants together.

The target customers include the members of the Mars society who are having a tough time adjusting to the new way of living. As they settle into their new homes and lifestyle, some may find difficulties with their mental health as well as their quality of life. Due to the lack of real-time communication, issues such as isolation, loneliness, sensory deprivation, interruptions to circadian rhythms, and anxiety can have an incredibly negative effect on members of society. Some additional stakeholders that may be relevant include the leaders of these colonies, as they are the ones funding this project. Researchers and engineers of NASA will also be target stakeholders for this project as well. The customers are looking for something that can engage them mentally and give them mental enrichment. They are also looking for something that can alleviate them from things like loneliness, homesickness, and depression. The stakeholders wish for an increase in morale as well as a decrease in anxiety and sensory deprivation.

Overall, the inhabitants of the Mars colonization are mentally struggling to adapt to living on a foreign planet. They are in dire need of mental stability and enrichment to improve their quality of life. They are in need of a sense of home to avoid negative feelings that can be detrimental if prolonged. The leaders of the colonization need peace and high morale for the safety and benefit of their civilization. Without these emotional and mental improvements, communication, teamwork, and emotional intelligence would plummet which would hinder any further research and discoveries on Mars. It is also important for NASA leaders that this be an affordable opportunity as the budget may need to be flexible for any Mars exploration projects and to maintain the day-to-day life of the colony. Here is a list of design requirements that the design solution that improves the mental health of users must fulfill:

- The design shall improve the overall mental health of the members of the Mars civilization by at least 5%
- The design shall be automated
- The design shall improve productivity
- The design shall withstand low atmospheric pressure
- The design shall withstand normal living conditions on the Mars base
- The design shall bring no harm to the user
- The design shall be automated

Here is a list of design criteria that a successful design solution must fulfill:

- Low cost: costs less than seventy-five dollars
- Low Maintenance
- Brings a sense of home to users

- Enables interaction between itself and the user
- Durability
- Low Weight

In this document, the design that improves the inhabitants' mental state, along with the other requirements, will be explored in order to bring a high-functioning, healthy mentality to the Mars colony. To begin, the design will be introduced– showcasing the many functions that will bring mental stability and happiness to the inhabitants of Mars. Then, the materials needed will be expressed in order to begin prototyping for the design. In addition, the original design choices will be included to convey the astute functionality of the final design that will bring the inhabitants to optimum working conditions. The developed AHP table, Table 1, portrays the importance of criteria in order of priority from least to most important as follows; low cost, low weight, durability, human interaction, sustainability, home feeling. To clear some vagueness, sustainability is in reference to how recyclable the exterior is of the robot and how long the battery can/will last. Baymax is utilizing balsa wood and plywood is best as it can be constructed from recycled materials available on mars.

Table 1: AHP Comparison Table of Different Criteria

Criteria	Low cost	Sustainable	Home feeling	Human interaction	Durability	Low weight	Total	Weight
Home feeling	6	6/5	1	6/4	2	6/2	14.7	.286
Sustainable	5	1	5/6	5/4	5/3	5/2	12.25	.238
Human interaction	4	4/5	2/3	1	4/3	4/2	9.8	.190
Durability	3	3/5	1/2	3/4	1	3/2	7.35	.143
Low weight	2	2/5	1/3	2/4	2/3	1	4.9	.095
Low cost	1	1/5	1/6	1/4	1/3	1/2	2.45	.048
							51.45	

Low Cost < Low weight < Durability < Human interaction < Sustainable < Home Feeling

Rationale: 1-5

1. Does not fulfill the criteria
2. Minimally fulfills the criteria
3. Fulfills the criteria adequately
4. Fulfills the criteria almost completely
5. Fulfills the criteria completely

Detailed Design Description

The final design is a cube-shaped robot with wheels. The robot's body is to be 6 inches in length, 6 inches in width, and 6 inches in height, for both the design and prototype, with its body composed of recycled plastics and building supplies, as shown in **Figure 1** below, collected around the Mars colony. Its primary function is to be an everyday companion for the Mars inhabitants. As stated above, some inhabitants seem to struggle daily with mental health, and in turn their mood management, so Baymax helps console its users with daily inspirational or kind messages, aromatherapy, and musical therapy. The inspirational messages are provided on the robot's LED screen, which is 5.14 inches in length and 2.7 inches in height, as seen in the dimensions of **Figure 5**, on the front of the robot. The aromatherapy and musical therapy are initiated off of a red and a green button, indicating a positive and negative mood, which then releases scents depending on which button is pressed, off the top of the robot out of a diffuser, and plays music, again depending on which button is pressed, out of speakers on the left and right sides of the robot. In the event that the user is injured or is experiencing pain, Baymax also has medical supplies, such as bandaids, ibuprofen, and Neosporin ointment, located in a medical drawer, with a length of 5.85 inches, a width of 2 inches, and a height of 2.15 inches that is located on the top of the robot. The drawer is automatically popping open, using a photoresistor connected to a servo motor. When the user's hand comes towards the drawer, the shadow cast by the hand over the photoresistor will activate the drawer opening. The titanium arms move into position once the user is within 5 inches of it. In order for the robot to be there for the user with their daily needs, the distance sensor is also connected to the Hobby Gearmotor with wheels, making sure to stay within 3 feet of the user at all times. **Figure 1**, which gives a front, top, and side view, labels all of the features of the robot that have been mentioned above. **Figure 2**, **Figure 3**, and **Figure 4** give clearer front, back, and side views of the final design as well.

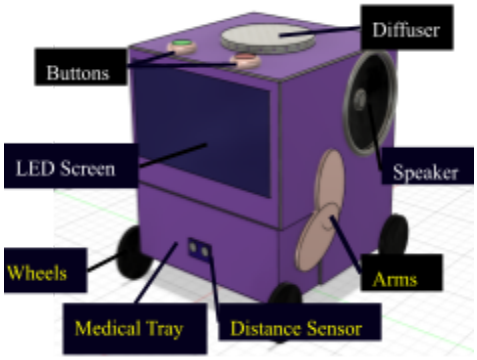


Figure 1. Labeled 3D CAD Model



Figure 2. Front View of CAD Model



Figure 3. Back View of CAD Model

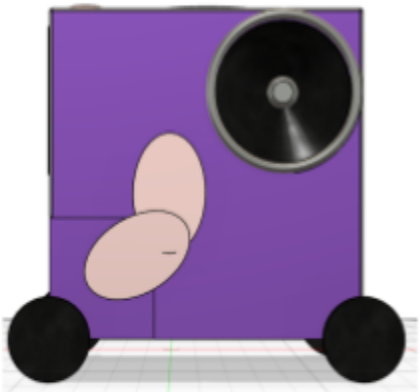


Figure 4. Side View of CAD Model

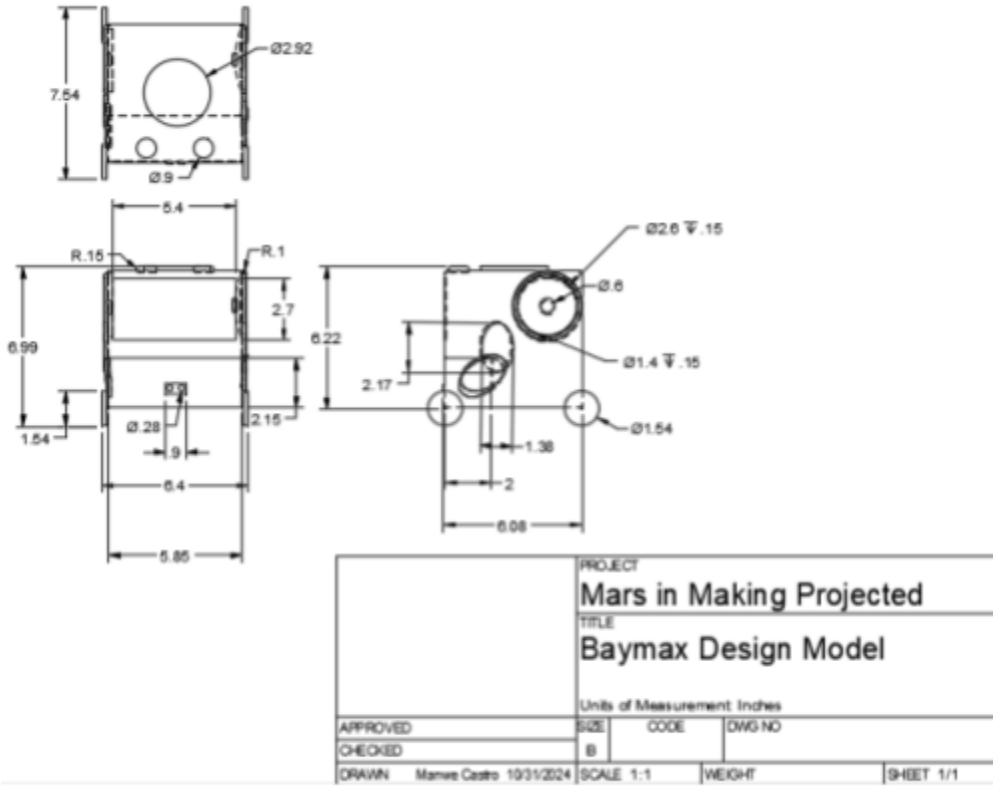


Figure 5. Orthographic Drawing with Dimensions

The design works to solve the mental health problems of inhabitants by using all the included features in a variety of ways. First, the robot is designed to automatically come up to users to perform its purpose. Once it has come into close enough proximity to the user it will activate the drawers and offer the user items that have been previously stored. It will then display a nice message for the user to read. This can make users feel appreciated and wanted as the robot seeks them and provides them with a companion. The new Mars colony most likely lacks a sense of community and inhabitants can feel isolated. The robot can stimulate inhabitants and make them feel less lonely in the Mars colony while away from family and friends.

For the prototype, the same dimensions of 6 inches by 6 inches by 6 inches from the original design were used. Instead of the recycled plastics and building supplies collected around

the Mars colony, it will be made ⅛” Balsa and plywood. The robot will have an LCD screen instead of an LED screen, with dimensions of 3.15 inches in length and 1.42 inches in height, and display simple messages such as “Good Morning” as shown in **Figure 6.** The medical drawer was moved to the top of the robot, still with 3 compartments, and opened by a small piece of wood connected to a servo motor. The input for that function will be a photoresistor on the top, as shown in **Figure 7,** and is determined by having a light value of less than 25 as shown in **Appendix A**. A Hobby Gearmotor with wheels will still be a function of the prototype, still automated by the distance away from the distance sensor, as shown in **Table 2**. The overall function of the robot will continue to be an everyday companion for Mars inhabitants.



Figure 6: Final Prototype Front View



Figure 7: Prototype Top View

Table 2: Prototype Budget and Materials List

QUANTITY	MATERIAL	PRICE
1	Servo motors	$2.00
2	Hobby gear motors w/ wheels	$9.00
2	103m Jiffy Clips	$0.20
2	3” diameter O Ring	$0.20
4	Rubber Grommet	$0.40
1	Small Flat bracket	$0.50
1	Potentiometer	$1.00
4	Small hinges	$6.00

1	photoresistor	$1.50
1	LCD screen	$6.00
1	Ultrasonic sensor	$2.00
2	9V battery	$5.00
1	Breadboard	$2.00
1	L29D 16-Pin Motor Driver Controller	$1.00
2 sq ft	⅛” Balsa Wood	$6.00
2 sq ft	⅛” Plywood	$1.40
TOTAL		$44.20

The automated features of the robot include the drawer lid for the medical drawer and the wheels that move the robot to follow the user, as shown in the highlighted lettering in Figure 1. The lid will move using a small piece of wood connected to a servo motor with its automated input being the photoresistor. and a distance sensor so that the robot knows to move after the user has moved a certain distance toward the medical storage area. For instance, **Diagram 1**. shows that once the light is calculated from the photoresistor when the light is less than 25, the drawer will be opened for 5 seconds. As for the wheels, an ultrasonic sensor will calculate the distance between the user and the robot. If the user is more than 5 inches away, the robot will move forward using hobby gear motors and wheels. After considering the critique given, it has been decided that the arms of the robot will no longer extend. Instead, the focus will be on making the robot move to follow the user, creating a feeling of a companion further.

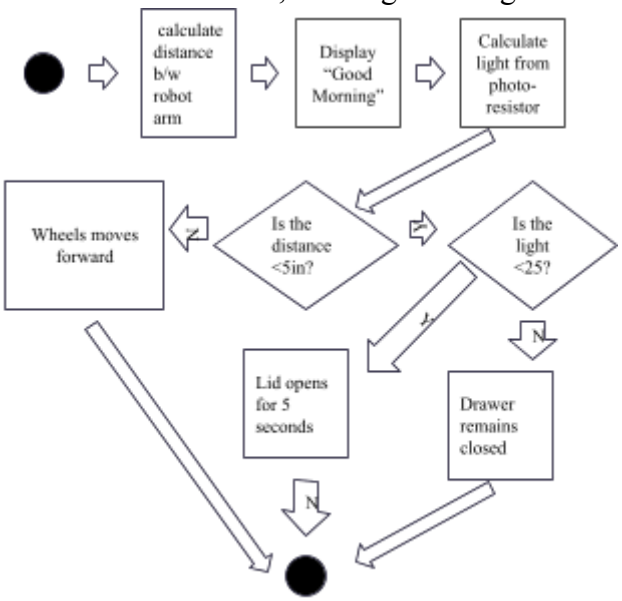


Diagram 1: Activity Diagram

This design is unique because it incorporates aromatherapy, anthropomorphism, and music therapy to better the health of the users. Other products in the market to increase the mental health of users such as stuffed animals act as comfort objects using anthropomorphism; however, the design relies solely on those object-attachment tendencies. Rather than bringing more aspects to improve mental health into consideration such as medications that must be taken, a stuffed animal does not function to actively help the user. By having a robot with a medicinal drawer always nearby, the user will be supported in their day-to-day life. For instance, an elder who needs motivation to take their daily medications would now have a pet robot there to support and encourage them to maintain their health. As well, the user can interact with the robot to play music, diffuse scents, and read little messages the robot brings. This is sustainable for the environment on Mars as the body can easily be built from recycled materials such as scrap wood, abandoned wires, and partially used batteries. This will allow the robot to thrive in the dome and continue to support users as they move from Earth to Mars for years to come.

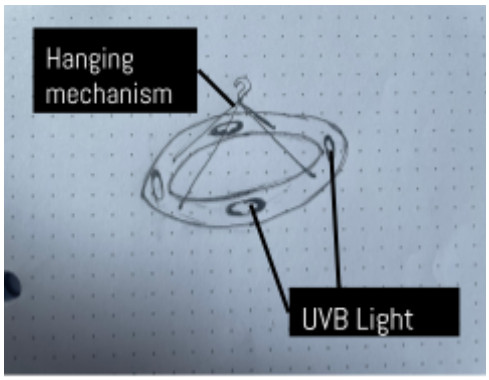


Figure 8. Daylight Simulator Sketch

Throughout the design process, multiple ideas were considered. “Daylight Simulator” uses models similar to a circular chandelier as seen in **Figure 8.** with four UVB light bulbs. This is important because UVB light emits a spectrum of ultraviolet light with wavelengths ranging from 290–320 nanometers so that the user can get the proper amount of vitamin D. The light will be set on a timer and have a dimmer to simulate dusk, day, and dawn. This model would help maintain circadian rhythm by simulating the Earth’s daylight cycle.



Figure 9. Baymax Design Sketch

“Baymax”, inspired by the healthcare companion from Big Hero Six, is modeled in **Figure 9.** as a small robot with an LED screen in the front to show a face with messages (good morning, good night, have a good day). It will have a medical tray full of bandaids, Neosporin, and alcohol wipes that lift towards the user. The robot will also have an aromatherapy humidifier located at the top of its ‘head’ and speakers on its sides to play relaxing music/audio for sound therapy.. It will have four wheels so that it can follow or sit stationary as the user pleases.

“Mood Button” in **Figure 10.** is a futuristic watch that wraps around the forearm and screens different buttons that represent a

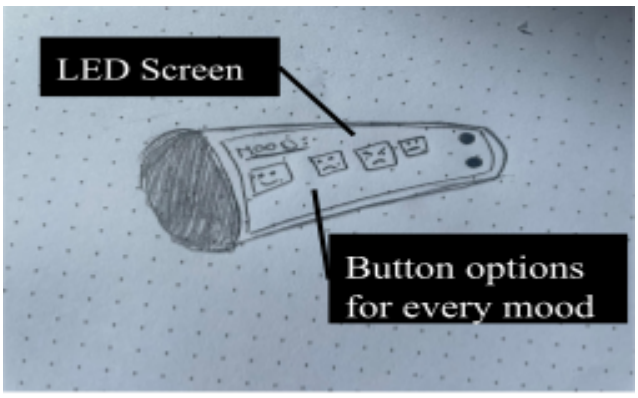


Figure 10. Mood Button Sketch

specific mood. The user will be able to press the emotion they are feeling and the smart band will activate headphones or speakers and humidifiers to play specific music/audio and emit scents to alleviate the user's mood. The "Mood Button" can alert friends and family of their mood which will improve how teams interact with each other.

Daylight Simulator: As conveyed in **Table 3,** the home feeling is a 3 because while it gives the person a feeling of the sun moving up and down as it did on Earth, it does not especially bring feelings of home. The sustainability is a 3 because the light bulbs would use nuclear power– not super maintainable for Mars. Human interaction is a 1 because there is no human interaction with the light. The durability is a 3.5 because light bulbs are pretty durable in their longevity but light bulbs themselves are very fragile. For low weight, the daylight simulator gets a 5 because light bulbs are light. For cost, it gets a 2 because the light bulbs that give off vitamin D are very expensive.

"Baymax": The home feeling is a 5 because the design will be a comfort object that will give the colonists a feeling of home to a greatly large extent. The durability is a 3 because it should last a good amount of time as a robot. However, its mechanical parts may struggle with the atmosphere on Mars. The maintainability is a 5 because it will use batteries which will be easy to transport from Earth to Mars. In addition, solar power could be used to charge the battery, so there are many different options that make it maintainable. Interactive is a 5 because it will be a personal assistant that is always there with the user and can help with medical assistance. Low weight is a 2 because all the mechanical parts will be pretty heavy which is not ideal. Low cost is a 2 because all the parts to make it may be expensive since it has many different functions.

Mood Button: The home feeling is a 4 as it has the possibility to simulate the scents and sounds commonly heard from their home. The durability is a 4 as it is meant to last a long time for each individual person. The maintainability is a 3 because similar to the Baymax idea, it could use batteries or be solar-charged. This makes it decently maintainable. Human interaction is a 5 as it directly communicates with other members of the colony. The low weight is a 4 as it is intended to be worn on the wrist. The low cost is a 2 as some of the components needed for them may be a bit expensive.

Table 3, the decision matrix, determined that "Baymax" is the best design to fulfill the project's goal because it requires user interaction, promotes a homey feeling, and is very maintainable in comparison to "Daylight Simulator" and "Mood Button" where interactiveness and maintenance are lower respectively. Though the "Daylight Simulator" would improve physical well-being by maintaining the user's circadian rhythm by emitting synthetic vitamin D, it wouldn't have promoted mental stimulation. The "Mood Button" is still thought to be a good idea but it lacked the requirement of a motor and also didn't effectively fulfill the criteria as well as "Baymax".

After this decision, the design for "Baymax" was revised to include features similar to "Mood Button". Speakers and a humidifier will be added to "Baymax" since aroma therapy and musical/sound therapy have been proven to improve moods and promote relaxation. The development of the arms was brainstormed from being high in dexterity to a more minimalistic approach. A motorized telescopic arm was added along with an 'elbowed' arm that moved only

vertically. The robot was then simplified to the final design with just an ‘elbowed’ arm so that the motion of the Medical Tray will be more consistent which is shown below in **Figure 10**.

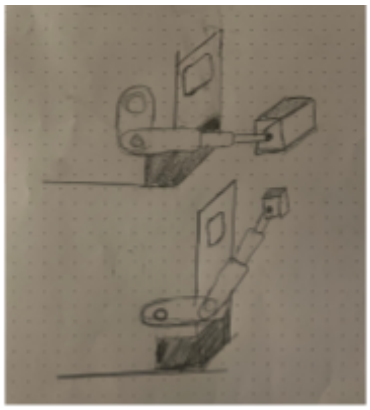


Figure 10. Second Revision of ‘Baymax’

To strengthen the rationale for Baymax’s design and why it was chosen in the Decision Matrix, **Table 3**, different therapies are integrated that are scientifically-backed methods to address Mars inhabitants' mental health needs through sensory engagement. The addition of aromatherapy is supported by research indicating that essential oils can positively impact mental health. According to Johns Hopkins Medicine, certain scents, such as lavender and chamomile, can alleviate stress and anxiety by promoting relaxation and reducing cortisol levels, a key benefit for Mars colonists who experience heightened isolation. These scents, when diffused by “Baymax” based on the user's emotional state, provide a familiar, comforting atmosphere, enhancing the "home feeling" criterion in our design matrix. Additionally, research on anthropomorphic devices shows they significantly reduce feelings of loneliness by simulating human-like interactions. A study in the National Library of Medicine emphasizes that personal voice assistants and anthropomorphic features improve social connection, particularly in isolated environments. Baymax’s friendly appearance and interaction model mirror these principles, acting as a psychological buffer for the Mars inhabitants by encouraging routine engagement and empathy, which enhances their mental resilience in isolation. Finally, the inclusion of auditory therapy aligns with findings in Taylor & Francis Online, which highlight music's therapeutic effects on mental well-being. By pairing aromatherapy with calming sounds, Baymax effectively leverages multisensory engagement to reduce the inhabitants’ stress and improve their overall morale, directly supporting our goal of a 5% mental health improvement and ensuring Baymax’s lasting impact on user well-being.

The face of the design includes an LCD screen centered at the top with an ultrasonic sensor at the bottom. The user should be able to see the messages from the robot as they face it, so the LCD screen was put on the face. Since the user should be in front of the robot, the ultrasonic sensor was put on the bottom of the front of the screen so that it is out of the way but still able to calculate the distance between the user and the robot. On the sides of the robots, there are powered wheels in the back and regular wheels in the front. The powered wheels were put in the back for aesthetic purposes, but they also allowed room for the ultrasonic sensors and LCD wires on the inside of the robot. The top of the robot features a lid that opens to a medical box along with a photoresistor. The lid was put on the top for the user’s convenience since it is the easiest place to reach. The lid opens based on the light read from the photoresistor, so the best place for the photoresistor is on top so that the lid will open based on how close the user’s hand is to the lid. The user’s hand will cast a shadow to the photoresistor which causes the lid to open.

Initially, the design included many features for customers to engage with the robot. Some of the ideas included aromatherapy, music therapy, and kind messages displayed on an LCD screen. The robot would be composed of a box shape and driven on wheels. It would have also included a drawer that could hold a variety of items ranging from first aid to fun snacks. This drawer was to be opened up by automated arms and to hand you the things inside it. The aromatherapy and musical therapy would be controlled by corresponding buttons to turn on and

off a diffuser releasing a nice scent and music of the user's choice. From the original idea, a lot of things changed in the final product. First off the concept of buttons was removed completely and the aromatherapy and speakers went with it. If this project were to be conducted in a real-life scenario it would include a function diffuser and speaker. For the sake of this project, it will only be modeled while our team builds its functioning motors. We stuck with the idea of a box robot on wheels and included the LCD screen displaying kind messages but instead, it would only display one message, “Good Morning”. We also kept the drawer and it was designed so that whenever the photoresistor is activated it would swing open for a user to grab whatever they want themselves instead of the robot handing it to them.

Table 3: Decision Matrix

Criteria	**Weights (%)**	**Daylight Simulator**		**Baymax**		**Mood Button**	
		Rating Factor	Weighted R.F.	Rating Factor	Weighted R.F.	Rating Factor	Weighted R.F.
Home feeling	**.286**	3	.858	5	1.287	4	1.144
Maintenance	**.238**	3	.714	5	1.19	3	0.714
Interactive	**.190**	1	.190	5	0.95	5	0.95
Durability	**.143**	3.5	.501	3	0.429	4	0.572
Low weight	**.095**	5	.475	2	0.19	5	0.475
Low cost	**.048**	2	.096	2	0.096	2	0.096
Total		2.83		4.142		3.951	

Design Performance

It was difficult to perform preliminary testing until the robot was completely together, so the first informal preliminary testing for this project was done on November 21 where the input sensors, the photoresistor, and the ultrasonic sensor, were tested. It was determined that day that the hobby gear motors weren't properly functioning as they'd have a delayed reaction or no reaction at all to the ultrasonic sensor. The team's initial assumption was that there wasn't enough battery power to turn the wheels since the LCD screen successfully completed the expected outcome. As for the second input, the photoresistor, a completed outcome was achieved and the servo motor opened the medical tray whenever a hand covered the photoresistor.

In an effort to fix the mechanics before acceptance tests needed to be completed on November 26, the team spent the weekend solving these issues. While trying to figure out the reason why the hobby gear motors weren't reacting to the distance sensor, it was determined that it was due to a lack of battery power and friction against the Balsa wood made it difficult for the wheels to turn, so the team sanded down the corner to properly outline the circumference of the wheel to prevent any friction as can be seen on the red outline on Figure 11. On November 24, the robot successfully completed all of our requirements after we informally tested it. The hobby gear motors reacted when the user wasn't 15 cm in proximity and the LCD screen read "Good Morning" after the distance sensor was activated; however, the servo motor did not open the medical tray when the user's hand covered the photoresistor.

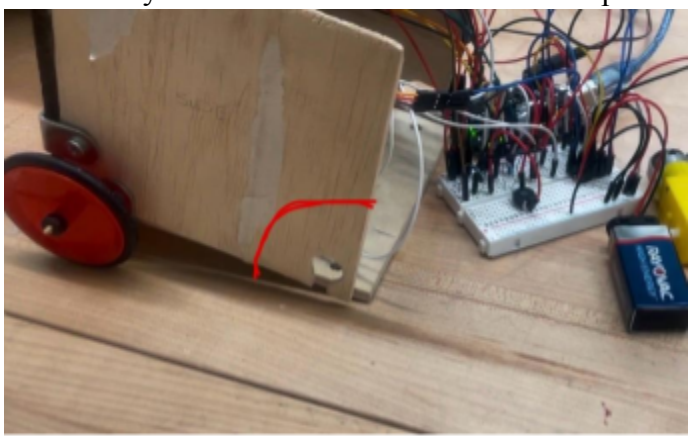


Figure 11: Friction against Wheel

Coming back to class on November 26, the day of acceptance testing, the team found that there was a wiring error that was made the days prior when the wires were being readjusted to add more battery power that caused the servo motor output to be disconnected from the photoresistor input. By the end of class on November 26, the robot had successfully passed all of the acceptance testing which can be seen in Appendix B along with the prerequisite and the scope of the experiment.

To begin how the robot successfully improved the mental health of the user through the messages it portrays on the screen. Right now, the screen portrays "Good morning" and rotates through automated faces but with further progression of this project, more comforting messages will be displayed. In general, it was observed that the user's initial facial expressions were neutral and afterward they were left smiling. Other features such as aromatherapy and sound

therapy are included on the robot and are scientifically supported to have the ability to alleviate stress and anxiety. These combined aspects create a sense of companionship and proximity-based messages boost morale among Mars inhabitants. By addressing mental health issues, “Baymax” allows inhabitants to focus on their daily tasks, improving productivity. Its automation ensures that users do not need to spend time manually interacting with the robot for essential functions, such as accessing medical supplies. The robot’s features are fully automated. The medical tray opens when the user’s hand covers the photoresistor and the wheels allow the robot to follow the user within a set distance of 15 cm. The ultrasonic sensor and photoresistor, combined with the servo motors and hobby gear motors, achieve this automation seamlessly. Also, the materials used for the prototype (balsa wood and plywood) replicate a design that could be constructed from recycled materials available on Mars, warranting Baymax’s ability to be adaptable to low-resource environments. Lastly, the robot’s design is low-weight and critical for resource-constrained environments; while the prototype uses lightweight materials, future iterations on Mars would use durable, lightweight composites.

The performance of this experiment during the class presentation wasn’t able to be fully completed due to the robot beginning to smoke. Baymax followed the user until it was within 15 cm of them; however, the presentation had to be quickly shut off due to smoke seeping from inside the robot. At that point, a video of the robot being successfully conducted was played as evidence that its functions were repeatable. After taking a look inside the prototype, it was seen that the rubber on the wires connecting the battery to the breadboard had begun to melt together. It was determined that a wire connected to the battery wasn’t properly soldered and that some of the exposed wiring began to heat up due to power not being able to flow through it quickly enough.

Conclusion

The project went well as all components were working properly. The drawers were able to open without issues when the photoresistor was activated. The robot was able to approach users without any issues and the LCD screen worked perfectly. The only issue is towards the end of creating our prototype some of the wires were not soldered properly which caused some smoking to happen when the robot was running. Though this was a problem and could cause some safety concerns, the design still functioned properly and worked just as we wanted it to.

The design creates value for stakeholders because our product increases efficiency and productivity for all inhabitant users. People under high stress tend to perform worse at various activities and this product is the solution for this. Inhabitants experience a lot of stress from the extreme move to Mars and the loneliness and isolation they experience being away from their friends and family. The product is made to reduce this stress and relieve the isolated and lonely feelings inhabitants experience. This creates value not only for inhabitants looking for something to solve these mental health problems but also for stakeholders as it can create a better environment for all Mars inhabitants. With a better overall mood throughout the colony, everybody will be working more productively.

The design is also a sustainable solution because it is completely recyclable. It is made out of materials that can be reused, like all of the electrical components. Also, all the other components are made of wood and that makes it fully recyclable which is extremely beneficial for the Mars colony as they do not need outside resources to continue manufacturing it.

If the robot was actually given to the customer one thing that would be improved is the robot's ability to follow the user. In the current design, the robot is only able to move forward and in a real-world situation where the robot needs to approach the user wherever they are it would be helpful for it to be able to change directions. Another thing that would be added would be musical therapy and aromatherapy as included in our original model. To implement this the robot would need speakers and a diffuser that are compatible with Arduino. The builders would need to cut adequate holes in the robot to include this and maybe even make it bigger so all the new components would be able to fit.

If the robot was mass-produced then instead of wood for the robot plastic could be more beneficial to use. Also, a bigger battery could be helpful to keep the robot on for longer periods of time before needing to be charged. The wire management should also be cleaned up as if it is disorganized it causes the risk of a short circuit or a fire. Generally, mass production of this product shouldn't be that difficult as there aren't too many components and the design is relatively simple.

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Appendix A: Arduino Code and Circuits

```
#include <Servo.h>  //includes the code in the servo
#include <LiquidCrystal.h>

Servo hubert;  //top horizontal

//distance sensor
int trigPin = 2, echoPin = 3;
long distance, duration;

//top left gear
int motorpin5 = 10, motorpin6 = 11;
//bottom left gear
int motorpin7 = 12, motorpin8 = 13;
//lcd
char myTEXT1[] = "GOOD";
char myTEXT2[] = "MORNING!";
char myTEXT3[] = "0";
char myTEXT4[] = "-";
char myTEXT5[] = "^";
char myTEXT6[] = "PLACE HAND";
char myTEXT7[] = "ABOVE";
// LiquidCrystal lcd(RS,E,DB4,DB5,DB6,DB7):
LiquidCrystal lcd(5, 15, 16, 17, 18, 19);

int light;

void setup() {
  Serial.begin(9600);
  lcd.begin(16, 2);

  pinMode(echoPin, INPUT);
  pinMode(trigPin, OUTPUT);


  pinMode(motorpin5, OUTPUT);
  pinMode(motorpin6, OUTPUT);
  pinMode(motorpin7, OUTPUT);
  pinMode(motorpin8, OUTPUT);

  pinMode(8, OUTPUT);
  hubert.attach(8);
  hubert.write(90);

  Serial.println(light);
}
void loop() {
  light = analogRead(A0);

  //make trigger low
  digitalWrite(trigPin, LOW);
  delay(10);

  //make trigger signal(
  digitalWrite(trigPin, HIGH);
  delay(10);
  digitalWrite(trigPin, LOW);  //)
  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;


  Serial.println(distance);

  if (distance > 15) {

    digitalWrite(motorpin5, LOW);
    digitalWrite(motorpin6, HIGH);
    digitalWrite(motorpin7, LOW);
    digitalWrite(motorpin8, HIGH);

    lcd.clear();

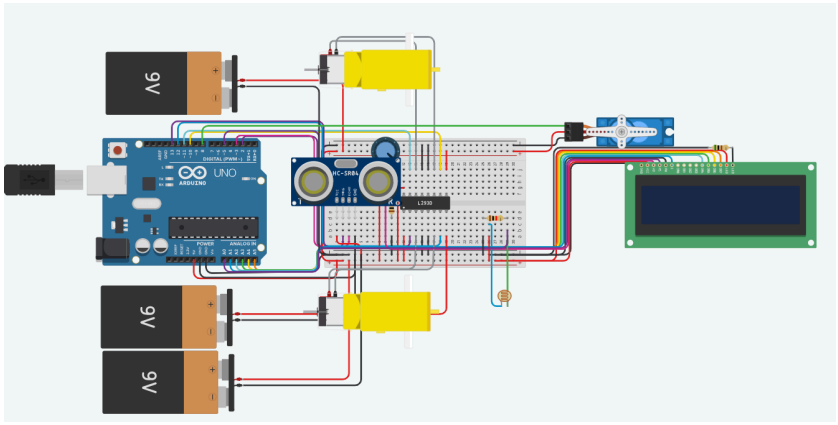
  } else {
    digitalWrite(motorpin5, LOW);
    digitalWrite(motorpin6, HIGH);
    digitalWrite(motorpin7, LOW);
    digitalWrite(motorpin8, HIGH);

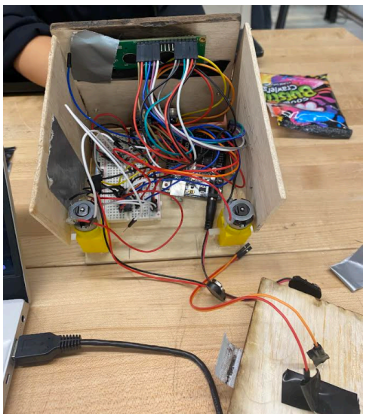
    lcd.clear();

  } else {
    digitalWrite(motorpin5, LOW);
    digitalWrite(motorpin6, LOW);
    digitalWrite(motorpin7, LOW);
    digitalWrite(motorpin8, LOW);

    if (light > 25) {
      hubert.write(90);
    } else {
      hubert.write(0);
      delay(5000);
      hubert.write(90);
    }
    //good morning
    lcd.setCursor(6, 0);
    lcd.print(myTEXT1);
    lcd.setCursor(5, 1);
    lcd.print(myTEXT2);
    delay(5000);
    lcd.clear();

    //place hand above
    lcd.setCursor(2, 0);
    lcd.print(myTEXT6);
    lcd.setCursor(5, 1);
    lcd.print(myTEXT7);
    delay(3000);
    lcd.clear();
  }
}
```





Appendix B: Acceptance test procedures and results.

Scope: This test verifies that the robot functions in a way that promotes the mental health of the user by helping the user receive medical equipment by opening a lid and remaining nearby.

The following requirements are included in this procedure:

1. Wheels move forward when the user is more than 15 cm away.
2. The lid to the medical tray moves up when there is no light.
3. Wheels stop when the user is less than 15 cm away.
4. LCD Screen turns on to read “Good morning” when the user is 15 cm away.

Prerequisites:

- 2 functioning hobby gear motors
- 1 functioning servo motor
- 1 functioning photoresistor
- 1 functioning ultrasonic sensor

Test Procedure:
11/21/2024

Step	Instructions	Expected Outcome	Requirement	Pass	Fail
1	Move the robot away from user	Robot will move forward until its within 15 cm of the user	Wheels move forward when the user is more than 15 cm away	☐	☑
2	Hold hand in the front side direction of the robot	The robot will stop when the user’s hand is 15 cm away	Wheels stop when the user is less than 15 cm away	☐	☑
3	Hold hand in the front side direction of the robot	The robot’s screen will turn on when the user’s hand is 15 cm away	The robot’s screen will read “Good morning”	☑	☐
4	Hover hand on top of the robot	Medical tray will open	The lid to the medical tray moves up when there is no light	☑	☐
5	Hover hand on top of the robot	Medical tray will open	The lid to the medical tray moves up when there is no light	☑	☐

11/24/2024

Step	Instructions	Expected Outcome	Requirement	Pass	Fail
1	Move the robot away from user	Robot will move forward until its within 15 cm of the user	Wheels move forward when the user is more than 15 cm away	☑	☐
2	Hold hand in the front side direction of the robot	The robot will stop when the user's hand is 15 cm away	Wheels stop when the user is less than 15 cm away	☑	☐
3	Hold hand in the front side direction of the robot	The robot's screen will turn on	The robot's screen will read "Good morning"	☑	☐
4	Hover hand on top of the robot	Medical tray will open	The lid to the medical tray moves up when there is no light	☐	☑
5	Hover hand on top of the robot	Medical tray will open	The lid to the medical tray moves up when there is no light	☑	☐

11/26/2024

Step	Instructions	Expected Outcome	Requirement	Pass	Fail
1	Move the robot away from user	Robot will move forward until its within 15 cm of the user	Wheels move forward when the user is more than 15 cm away	☑	☐
2	Hold hand in the front side direction of the robot	The robot will stop when the user's hand is 15 cm away	Wheels stop when the user is less than 15 cm away	☑	☐
3	Hold hand in the front side direction of the robot	The robot's screen will turn on	The robot's screen will read "Good morning"	☑	☐

4	Hover hand on top of the robot	Medical tray will open	The lid to the medical tray moves up when there is no light	☑	☐
5	Hover hand on top of the robot	Medical tray will open	The lid to the medical tray moves up when there is no light	☑	☐

Appendix C: Team member contributions

Team Member	Contributions to Report	Contributions to Prototype
Manwe Castro	Design Description, CAD Model, Orthographic Drawing with Dimensions, Appendix A,	Built the tinkercad circuit and code, Built the physical circuit, Coded the Arduino, Drawing on the back of the prototype, Troubleshooting issues, Building the exterior of the prototype,
Laura Londrie	Activity diagram, wrote a couple paragraphs in different parts, edited ,contributed to Appendix B	Troubleshooting, soldering, edited code a bit, circuit-building slightly, and morale
Nina Suberville	Wrote the Design Performance and contributed to Appendix B, helped label Figures and Tables	Laser-cutting, calculating the measurements of the prototype, building the exterior of the prototype, morale, developed the structure of the medical tray,
Krishna Gade	Design Performance, Detailed Design Description, Conclusion	Woodwork, cutting the prototype to make components fit, building exterior of prototype, helped laser cutting,